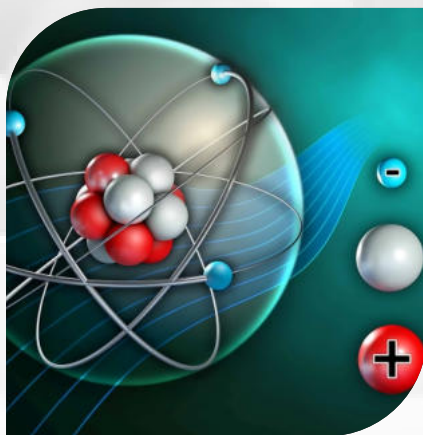


UNIT 19

ATOMIC STRUCTURE



ATOM

The fundamental unit of matter, consisting of a nucleus containing protons and neutrons, surrounded by electrons in energy levels, representing a specific chemical element.

An atom is composed of two main parts:

1. Nucleus: The central core of the atom containing protons (positively charged particles) and neutrons (uncharged particles), concentrated mass, and responsible for the atom's identity.

2. Electron Cloud: The region surrounding the nucleus where electrons (negatively charged particles) orbit in energy levels, determining the atom's reactivity and interactions.

ATOM NUCLEUS

The central core of an atom, composed of positively charged protons and neutrally charged neutrons, housing most of the atom's mass and determining its chemical properties.

GEIGER AND MARSDEN A – SCATTERING EXPERIMENT

The Geiger-Marsden alpha scattering experiment, also known as the gold foil experiment, was conducted in 1909 by Hans Geiger and Ernest Marsden under the guidance of Ernest Rutherford. The experiment played a crucial role in shaping our understanding of the structure of the atom.

EXPERIMENTAL SETUP

In the experiment, a thin sheet of gold foil was placed in the path of a beam of alpha particles, which are positively charged particles emitted by a radioactive source. The alpha particles were directed towards the gold foil, and a circular detector screen surrounded the foil.

EXPECTED OUTCOME

Based on the prevailing model of the atom at the time, which was J.J. Thomson's "plum pudding" model, it was expected that the alpha particles would pass through the gold foil with minimal deflection. According to this model, atoms were thought to be composed of a diffuse cloud of positive charge with embedded negatively charged electrons.

ACTUAL OBSERVATIONS

However, the experiment yielded surprising results. While most alpha particles passed through the foil as expected, some of them experienced significant deflection, and a small fraction even bounced back in the direction from which they came. This observation was highly unexpected and defied the predictions of the plum pudding model.

SIGNIFICANCE

The Geiger-Marsden experiment revolutionized our understanding of atomic structure, leading to the modern concept of the atom with a nucleus containing protons and neutrons, and electron orbits around it. This model forms the foundation of nuclear physics and atomic theory today.

PROTONS, NEUTRONS

COMPOSITION OF ATOM

An atom is composed of subatomic particles:

1. Protons: Positively charged particles found in the nucleus, contributing to the atom's mass and identity. Each proton has a charge of +1.

2. Neutrons: Neutrally charged particles located in the nucleus alongside protons. They add mass to the nucleus without affecting the atom's charge.

3. Electrons: Negatively charged particles orbiting the nucleus in energy levels or electron shells. They contribute to the atom's size and reactivity. Each electron has a charge of -1.

The nucleus, containing protons and neutrons, forms the atom's central core, while the electron cloud surrounds it. The number of protons determines the element's identity (atomic number), while the sum of protons and neutrons gives the atomic mass.

ELEMENTS

An element is a substance composed of atoms with the same number of protons, defining its unique chemical properties and identified by its atomic symbol.

ATOMIC NUMBER

The atomic number is the count of protons in an atom's nucleus, determining the element's identity and its place on the periodic table.

ATOMIC MASS

An element is a substance composed of atoms with the same number of protons, defining its unique chemical properties and identified by its atomic symbol.

ELEMENT	ATOMIC NUMBER	ATOMIC MASS	PROTONS	NEUTRONS	ELECTRONS	SYMBOL
Hydrogen	1	1	1	0	1	H
Helium	2	4	2	2	2	He
Lithium	3	7	3	3	3	Li
Beryllium	4	9	4	4	4	Be
Boron	5	11	5	5	5	B
Carbon	6	12	6	6	6	C
Nitrogen	7	14	7	7	7	N
Oxygen	8	16	8	8	8	O

NUCLIDES

Nuclides are distinct atomic species characterized by the specific number of protons (giving the element's identity) and neutrons in their nuclei. Each nuclide has a unique atomic number and mass number, representing the total number of protons and neutrons in the nucleus.

Nuclides may have different isotopes with varying neutron counts while retaining the same atomic number, belonging to the same element but having different atomic masses.

ISOTOPES

Isotopes are atoms of the same element with the same number of protons (same atomic number) but different numbers of neutrons (different atomic masses), resulting in distinct atomic weights.

ISOTOPES OF HYDROGEN

Nuclides are distinct atomic species characterized by the specific number of protons (giving the element's identity) and neutrons in their nuclei. Each nuclide has a unique atomic number and mass number, representing the total number of protons and neutrons in the nucleus.

Nuclides may have different isotopes with varying neutron counts while retaining the same atomic number, belonging to the same element but having different atomic masses.

ISOTOPE	PROTONS	NEUTRONS	ATOMIC MASS
Hydrogen-1	1	0	1
Hydrogen-2	1	1	2
Hydrogen-3	1	2	3

Hydrogen-1 (often just called **Hydrogen**) has one proton and no neutrons. Hydrogen-2, also known as **Deuterium**, has one proton and one neutron. Hydrogen-3, known as **Tritium**, has one proton and two neutrons.